© The Hong Kong Polytechnic University

THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF LAND SURVEYING AND GEO-INFORMATICS

Programme	**:**	34478 BSc (Hons) in Land Surveying and Geo-Informatics 34478-SY BSc (Hons) in Land Surveying and Geo-Informatics 34480-FYL BSc (Hons) in Land Surveying and Geo-Informatics 34379 HD in Land Surveying and Geo-Informatics			
Subject Title	**:**	Geospatial Database Management and Design			
Subject Code	**:**	LSGI3245	**Session**	**:**	Semester 1, 2023/24
Date	**:**	9 December 2023	**Time**	**:**	12:30 – 14:30
Time Allowed	**:**	2 hours	**Subject Examiner(s)**	**:**	Dr CAO Rui (LSGI)

This question paper has a total of 6 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates: This paper has **TWO** sections, A and B

Section A: Answer **ALL** questions.

Section B: Answer any **ONE** question.

Total marks = 100.

Available from Invigilator: Nil

…/2

DO NOT TURN OVER THE PAGE UNTIL YOU ARE TOLD TO DO SO

© The Hong Kong Polytechnic University

Section A: Answer **ALL** questions.

Question 1 (True/False)

© The Hong Kong Polytechnic University

Question 3

What are the basic components of an entity-relationship (ER) model? Create an example showcasing these components and illustrating how they help to represent and organize relationships. (10 marks)

Question 4.

Illustrate the concepts of the primary key and foreign key in the relational data model using examples (i.e., tables with columns and rows). (10 marks)

Question 5

What are the main differences between a relational database management system (RDBMS) and a spatial database management system (SDBMS)? Name THREE differences and use 1-2 sentences to describe each point. (10 marks)

© The Hong Kong Polytechnic University

Question 6

Suppose we have three tables as follows. The ***stations*** table and ***blocks*** table store the bus stations as *points* and urban blocks as *polygons*, respectively; and their spatial relationship is shown in

Figure 1. The ***lines*** table provides details of the bus lines.

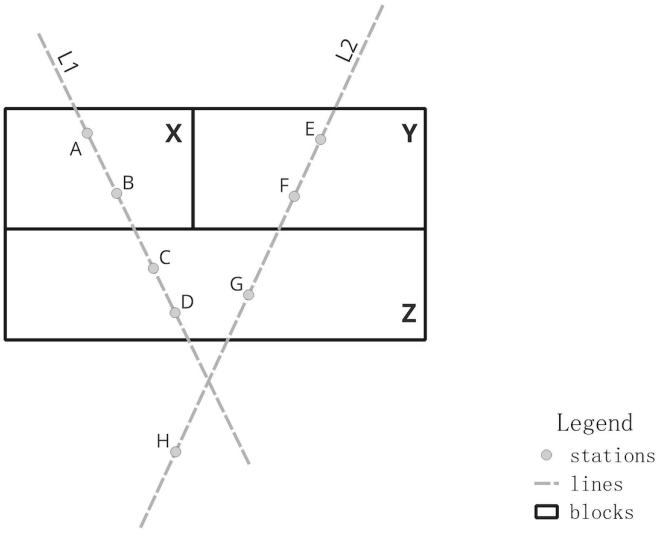


Figure 1: The map of the bus stations, bus lines, and blocks.

Table 1: **stations**

name (char)	**line_name (char)**	**num_entrances (int)**	**geom (geometry)**
A	L1	3	POINT1
B	L1	3	POINT2
C	L1	5	POINT3
D	L1	4	POINT4
E	L2	2	POINT5
F	L2	3	POINT6
G	L2	4	POINT7
H	L2	5	POINT8

Table 2: **lines** (length: *km*)

name (char)	**length (numeric)**	**open_year (int)**
L1	10	2009
L2	12	2016
L3	8	2013
L4	15	2018

Table 3: **blocks (**population: 10^4**)**

name (char)	**Population (numeric)**	**geom (geometry)**
X	2	POLYGON1
Y	3	POLYGON2
Z	6	POLYGON3

…/5

© The Hong Kong Polytechnic University

Based on these tables, we can perform spatial and non-spatial queries to find the information we want. For example, for the SQL expression "SELECT name FROM blocks", we can retrieve the following resulting relation:

name
X
Y
Z

Write down your answer in the form of a relation similar to the above example, for the following SQL expressions: (Note: ***ST_Within***(geometry *A*, geometry *B*) returns TRUE if geometry *A* is within geometry *B*.)

(a)
```
SELECT * FROM lines
WHERE length >= 10 AND open_year < 2018
ORDER BY length DESC, open_year ASC;
```
(5 marks)

(b)
```
SELECT lines.name, SUM(num_entrances) AS total_entrances
FROM lines LEFT OUTER JOIN stations ON lines.name = stations.line_name
GROUP BY lines.name ORDER BY lines.name;
```
(5 marks)

(c)
```
SELECT stations.name AS station_name, blocks.name AS block_name
FROM stations JOIN blocks ON ST_Within(stations.geom, blocks.geom)
WHERE blocks.name = 'Z'
ORDER BY stations.name, blocks.name;
```
(5 marks)

(d)
```
SELECT lines.name, SUM(population) AS served_population
FROM lines JOIN stations ON lines.name = stations.line_name
JOIN blocks ON ST_Within(stations.geom, blocks.geom)
GROUP BY lines.name ORDER BY served_population DESC;
```
(5 marks)

- END OF SECTION A -

© The Hong Kong Polytechnic University

Section B: Answer any **ONE** question.

Question 7

What are the performance bottlenecks of a spatial DBMS? Name at least TWO bottlenecks and use 1-2 sentences to describe each point. (10 marks)

Question 8

Explain how can R-Tree spatial indexes help speed up spatial queries? Create an example to illustrate this. (10 marks)

- END OF SECTION B -

- END OF PAPER -